

Marc Benedí

PH.D. CANDIDATE · COMPUTER VISION & DEEP LEARNING

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Education

Visual Computing Group - Technische Universität München

Munich, DE

PH.D., COMPUTER VISION & DEEP LEARNING

January 2023 - Present

- Researching the generation of human avatars from monocular depth observations, aiming to advance the capabilities of real-time interactive systems and enhance user immersion in digital environments.
- Teach graduate-level lectures and seminars on topics related to computer vision and machine learning.
- Supervise undergraduate and graduate student projects, providing guidance and expertise to support innovative research and practical applications in computer vision.

Technische Universität München

Munich, DE

M.SC., COMPUTER SCIENCE

October 2019 - September 2022

- Specialised in Computer Vision, Machine Learning and Deep Learning.
- Experience in CUDA/GPU programming with focus on high performance computing.
- Thesis: "Learning Robust Correspondences for Relative Camera Pose Estimation".

University of Edinburgh

Edinburgh, UK

ERASMUS+ (EXCHANGE)

January 2018 - May 2018

- Focus on Artificial Intelligence and Advanced Software Engineering lectures.

Universitat Politècnica de Catalunya, Barcelona Tech

Barcelona, ES

B.SC., COMPUTER ENGINEERING

September 2014 - July 2018

- Specialised in Computer Science.
- Thesis: "Design of an environment for solving pseudo-Boolean optimization problems".

Experience

Visual Computing Group - Technische Universität München

Munich, DE

RESEARCH FELLOWSHIP

May 2021 - April 2022

- Fellowship awarded to outstanding Master students to allow them developing state-of-the-art models.
- Developed a Convolutional Neural Network (CNN) for estimating correspondences and perform robust optimization to reconstruct 3D rigid-motions.

Visual Computing Group - Technische Universität München

Munich, DE

DEVOPS/SYSTEMS ENGINEER

April 2020 - September 2022

- Re-designed the laboratory's infrastructure into a high-performance SLURM cluster, specifically tailored for advanced visual computing and AI research, significantly optimizing task scheduling and resource allocation.
- Designed and implemented a full-stack platform for managing the cluster's scheduling, storage, and node resources, using Vue.js, Python and Docker for containerization.
- Deployed and sustained key infrastructure technologies such as LDAP, NFS, and PXE, improving system integrity and performance, and ensuring high availability and scalability of computing resources critical to research continuity.

WARP - Autonomus Driving

Munich, DE

COMPUTER VISION INTERN

April 2020 - September 2020

- Implemented Simultaneous Localization and Mapping (SLAM) for an autonomous vehicle, integrating sensor odometry and LIDAR-generated point clouds to enhance navigation precision. Utilized C++, Point Cloud Library (PCL), Robot Operating System (ROS), Ceres Solver, and Google Cartographer for robust and efficient mapping and localization.

Helm Mobile

Barcelona, ES

SOFTWARE ENGINEER

September 2018 - September 2019

- Developed Android applications leveraging Kotlin, Dagger for dependency injection, and Retrofit for API integration; engineered backend systems using Kotlin and SQL; and designed web interfaces with Vue.js and JavaScript—all with a strong focus on coding standards and Clean Architecture to ensure high-quality, maintainable, and scalable code.

CERN - European Organization for Nuclear Research

Geneva, CH

INTERNSHIP - "SUMMER STUDENT"

June 2018 - September 2018

- Engineered and deployed a Kubernetes cluster for the BE-OP-LHC team at CERN, enhancing system scalability, availability, and response time by 118%.
- Developed Ansible scripts to automate cluster deployment across various machine groups, ensuring consistency and efficiency in operations.

- Derived a novel conversion for transforming Binary Decision Trees into Conjunctive Normal Forms, enhancing the efficiency of Boolean Satisfiability Problem (SAT) representations and solutions.

Ernst & Young

- Developed responsive and dynamic web applications by creating the front-end using HTML, CSS, JavaScript, and jQuery, and building the back-end system in Java.

Projects

Animating the Uncaptured: Humanoid Mesh Animation with Video Diffusion Models

- A method to generate realistic 4D animations from a single 3D humanoid mesh and text prompt using generative video models and SMPL motion optimization.

Learning Robust Correspondences Estimation

- An end-to-end model for pose estimation between pairs of RGB-D frames. The key idea is learning confidence scores for correspondences in a self-supervised manner, helping to filter outliers and improve pose accuracy.

End-to-end Sparse Correspondence Prediction for Relative Camera Pose Estimation

- A method for end-to-end learnable, differentiable relative pose registration of RGB-D frames. The method predicts sparse correspondences using probability heatmaps, which then are fed into differentiable weighted Procrustes to estimate the final transformation between the frames.

KinectFusion

- Implemented "KinectFusion: Real-Time Dense Surface Mapping and Tracking" for real-time reconstruction of large-scale indoor scenes from RGB-D frame using C++ and CUDA for high-performance.

Divergence-Free Shape Correspondence with Time Dependent Vector Fields

- Extended the paper "Divergence-Free Shape Correspondence by Deformation" to represent sequences of non-rigid deformations as a time dependent vector field.

Honors & Awards

HACKATHONS

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| 2016 | GFI Sponsor Award , HackUPC Crowdify , a platform to create mosaics in big events using smartphones. | Barcelona, ES |
| 2016 | 1st Place + LIDL Sponsor Award , FoodHacks - iTrash , an Android app and smart-container to track the litter thrown away and automatically add it to the shopping list. | Berlin, DE |
| 2016 | 1st Place , StirHack - hackworld , an Android app to discover and plan trips to hackathons. | Stirling, UK |
| 2013 | Finalist , MANGO Innovation LAB | Barcelona, ES |

DISTINCTIONS

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| 2020 | Distinction , Grade award for the lecture <i>3D Scanning & Motion Capture</i> - Technical University of Munich | Munich, DE |
| 2018 | Distinction , Grade award for the lecture <i>Reasoning Agents</i> - University of Edinburgh | Edinburgh, UK |
| 2018 | Distinction , Grade award for the lecture <i>Software Testing</i> - University of Edinburgh | Edinburgh, UK |
| 2017 | Distinction , Grade award for the lecture <i>Computer Security</i> - Universitat Politècnica de Catalunya | Barcelona, ES |
| 2017 | Distinction , Grade award for the lecture <i>Software Architecture</i> - Universitat Politècnica de Catalunya | Barcelona, ES |
| 2015 | Distinction , Grade award for the lecture <i>Operating Systems</i> - Universitat Politècnica de Catalunya | Barcelona, ES |

Languages

English	C1 (Full working proficiency)
German	A1 (Basic)
Spanish	Native
Catalan	Native